

Measurement of Functional Activities in Older Adults in the Community¹

R. I. Pfeffer, MD, T. T. Kurosaki, MS, C. H. Harrah, Jr., MA
J. M. Chance, MD, and S. Filos, RN²

Two measures of social function designed for community studies of normal aging and mild senile dementia were evaluated in 195 older adults who underwent neurological, cognitive, and affective assessment. An examining and a reviewing neurologist and a neurologically trained nurse independently rated each on a Scale of Functional Capacity. Interrater reliability was high (examining vs. reviewing neurologist, $r = .97$; examining neurologist vs. nurse, $\tau_b = .802$; $p < .001$ for both comparisons). Estimates correlated well with an established measure of social function and with results of cognitive tests. Alternate informants evaluated participants on the Functional Activities Questionnaire and the Instrumental Activities of Daily Living Scale. The Functional Activities Questionnaire was superior to the Instrumental Activities of Daily Living Scale in discriminating among functional levels and in predicting neurologist ratings and cognitive scores. Used alone as a diagnostic tool, the Functional Activities Questionnaire was more sensitive than the Instrumental Activities of Daily Living Scale (.85 vs. .57) and almost as specific (.81 vs. .92) in distinguishing between normal and demented individuals.

Key Words: Social function, Aging

THE past decade has witnessed a growing recognition of the need for an objective and direct measure of social function in older adults. Performance in everyday life is linked more directly to social independence than are tests of cognitive function, which may be subject to multiple influences, some evanescent, including set and motivation, affective status, educational background, and experiences peculiar to specific cultural or socioeconomic groups. Gallagher et al. (1980) advocate multidisciplinary and multidimensional assessment techniques that sample cognitive function, affective status, physical health, and functional competence. Operational definition of mental illness now requires a social dimension to assess the presence and severity of disease (Spitzer & Endicott, 1978). Can an individual "function" normally? Does he or she require direct assistance or environmental compensation?

An index of social function will be useful (a) in identification of persons who require assistance, (b) in serial measurement of response to therapy, and (c) in determining whether re-

ported or measured cognitive and affective changes signify functionally meaningful disease.

An additional heuristic purpose may be served: investigation of the relation between social skills and cognitive function. Are everyday social and occupational skills so overlearned that they are less sensitive measures of brain function than neuropsychological tests, or do they represent separate, but related, attributes of innate and learned cerebral capacity? Investigation of such issues will require measures of functional capacity that are defined only by social function and that are as sensitive in that sphere as cognitive measures used for comparison.

We describe the development of a social function scale for use in a retirement community. We also describe a Scale of Functional Capacity (or decline) for the assessment and comparison of normal and demented older adults and preliminary validation studies.

RATIONALE FOR DEVELOPMENT OF THE SCALE OF FUNCTIONAL CAPACITY

Because the opinion of experienced clinicians was to be used as one criterion for the level of an individual's functional capacity, an operational description of various levels was required. The proposed scale should be independent of socioeconomic status, level of formal education,

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² Department of Neurology, University of California at Irvine Medical Center, 101 City Drive South, Orange, CA 92668. Reprint requests should be addressed to Dr. Pfeffer.

and intelligence and should measure current social function, relative to an individual's native capacity. Although it was anticipated that functional capacity would correlate with formally measured cognitive function, cognitive function descriptors were intentionally omitted from, or given only secondary descriptive place in, definitions of all levels of definite impairment. Each successive level was selected to be distinct, progressively more severe, and broadly recognizable to a trained health professional as corresponding to a social "disposition," such as independence in the community, home care, nursing home, or hospital. To permit later comparison with measures of cognitive function, a subjective estimate of residual mental function was assigned to each ordinal level of function as an anchor point. The risk of subjectivity in partialling out contaminating physical disabilities and the risk of including the effects of depression were accepted. The scale utilized is reproduced as Table 1.

RATIONALE FOR DEVELOPMENT OF A SOCIAL FUNCTION QUESTIONNAIRE

In the retired person, highly variable and often limited environmental demands restrict any range of activities that may be considered to evaluate potential social competence. One's latent capacity may not be taxed by most activities of daily living. Many previously developed measures, such as the Plutchik scale (Plutchik et al., 1970), the Performance Test of Activities of Daily Living (Kuriansky & Gurland, 1976), Kleban's scale (Kleban et al., 1976), and the Stockton scale (Gilleard & Pattie, 1977), emphasize low-level activities more suitable for study of the institutional samples on which they were developed. Others, such as the Remington and Tyre Social Functioning Schedule (Remington & Tyre, 1979), describe behaviors more appropriate to a working population, rely heavily on subjective, if subtle, scoring by an examiner who does not know the participants, merge several types of functional activity in the same question (e.g., meal preparation and toilette), and confound interpretation (illness, depression) with level of function. The instrument that best describes activities common to retired older adults is the Instrumental Activities of Daily Living (IADL) Scale (Lawton & Brody, 1969). Even this instrument does not sample the more complex behaviors of older adults who live in the community.

In developing a new tool, we accepted Lawton and Brody's (1969) intuitively appealing concept of a hierarchy of relevant skills. Activities of daily living such as feeding, dressing, bathing, and going to the toilet are more elemental than instrumental activities, such as shopping, doing the laundry, and conduct of simple fiscal transactions. Comprehension of current events, some recreational or avocational activities, reading, and more complex fiscal skills, such as balancing a checkbook, probably represent higher levels of social function. Activities selected should be nearly universal among the older adults under study. Some overlap with activities and descriptions in the IADL Scale was accepted. The questionnaire should be sufficiently brief to minimize any distress that might be aroused in a spouse, friend, or relative by an extended recital of incapacity.

METHOD

The functional scale and social function instrument were evaluated in the course of a study to validate a senile dementia diagnostic instrument which is described in greater detail elsewhere (Pfeffer et al., 1981). In that study, 195 adults, aged 61 to 91 yr, who lived in a stable retirement community of 22,000 persons were evaluated. Participants were referred by 33 physicians as normal or mildly demented. Persons considered mildly demented needed not be under care for cognitive complaints. Because of the study purpose, validation of an instrument to distinguish normal older adults from those with very mild senile dementia of the Alzheimer type (SDAT), referrals of severe SDAT or of persons with dementia of other known cause, such as stroke, were discouraged. All participants completed a standard medical history questionnaire, three brief cognitive function tests, and a self-assessment depression questionnaire (the CES-D, Weissman et al., 1977). For comparison with the brief cognitive function tests used, the Mental Status Questionnaire (Kahn et al., 1960) was included. Participants were then examined by a nurse with extensive training in neurology and by an experienced neurologist. After this initial evaluation, laboratory data and opinions on mental status were solicited from hospitals and physicians and were available for 177 participants. A second experienced neurologist, blinded to the examining neurologist's conclusions, reviewed data from his examination and from family physicians and hospitals. Neurolo-

Table 1. A Tentative Scale of Functional Capacity Based on Estimate of Functional Status

Functional Capacity Level	Descriptive title	Description	Residual mental function estimate (%)
1	Normal	Fully independent with no restriction: No cognitive impairment. Either no assistance or advice is required in the 10 areas of daily activities, IADL, or activities of daily living (ADL), or this does not deviate from an established life-long pattern (e.g., utilization of an accountant or business advisor).	100
2	Questionably affected: uncertain status	Qualified independence: (a) participant or alternate informant report less skill or greater difficulty than formerly in 10 areas of daily activities or IADL, but independent of others. Only compensation, if any, is self-imposed (e.g., notes, reminders) or (b) study staff observe minor word-finding problems, minimal impairment of immediate recall or recent events, questionable problem with spatial orientation in unfamiliar surroundings, but normally oriented to person, place, and date (± 1 day).	90
3	Mildly affected	Definite but mild restriction in normal activities or independence: Another supervises or advises in two or more of 10 formerly mastered areas of daily activities, or one or more areas of IADL, but participant still carries out activity. Example: requires very explicit or written directions to drive out of neighborhood, guidance in balancing checkbook but still handles own money. Alternatively, significant difficulty with activity, but spouse tolerates performance they described as substandard. Probably could not be gainfully employed, but lives independently and normally with slight environmental modification or intervention, such as one to three visits per week from friend/relative or redistribution of spouse's responsibilities.	75
4	Moderately affected	Require assistance in IADL, but carry out a portion of this function (e.g., writes own checks or pays at supermarket but another balances check book, submits medical bills, drives them to appointments or accompanies on bus. More severely affected persons in this category need intermittent supervisor (two times weekly reminders) in ADL (e.g., coordinating items of apparel, missed areas in shaving or makeup) but perform the function. Requires active external intervention for "normal" home life, but takes care of basic personal needs.	60
5	Moderately severely affected	Half or more of IADL (e.g., shopping, laundry, handling money, making appointments, transport, dispensing medication) are performed by other. Require three times weekly assistance with toilette, although may participate in dressing and feed themselves. Most are spontaneously ambulatory. Requires major readjustments of family routine, adult daycare, or variable degrees of attendant assistance; they may be placed in nursing homes.	40
6	Severely affected	Requires major, daily assistance in ADL (dressing, feeding, using toilet) and total assistance with IADL. Any ambulation with assistance. Initiates almost no activity other than for feeding, toilet, comfort. Institutional care is usual.	25
7	Severely incapacitated; helpless	Totally dependent. Mute, wheelchair, or bedfast. Almost all are institutionalized and require a higher level of nursing care than available outside most acute/rehabilitation hospitals.	10

gists and the nurse independently assigned each participant a score on the Scale of Functional Capacity and assigned a diagnosis (normal, depression only, senile dementia, senile dementia and depression, or other dementing illness) according to previously defined operational criteria. In case of doubt, social (dys)function was the final arbiter between normal and abnormal classifications. A mental function "index" representing a weighted average of scores on the Mini-Mental State Exam (Folstein et al., 1975), Symbol Digit Modalities test (Smith, 1973), and subtest B of the Raven Colored Matrices

(Raven, 1956) was derived by a discriminant function analysis, utilizing consensus diagnosis of the two neurologists as outcome. Depressed individuals and those of uncertain status were reevaluated by further neuropsychological tests before reaching the consensus diagnosis.

Each participant assessed his/her own functional ability on eight tasks such as making out business papers, travel, shopping, and hobbies as "normal now," "can't do or difficult," or "never did." With the participant's consent, an alternate informant, such as a spouse, child, or close friend, was asked to complete two mea-

asures of social function describing each participant. The IADL scale was utilized as originally published (Lawton & Brody, 1969), except that the phrase "Can your spouse... (do)..." in questions 1 to 3, 6, and 7 was modified to "Does your spouse..." (do various activities without help, with some help, etc.). Lawton and Brody do not recommend specific weights for levels of independence. For each IADL activity in the present study, independence was assigned a score of 1; dependent status, a score of 3; intermediate status, 2.

Our own instrument, the Functional Activities Questionnaire (FAQ), presented a forced choice among levels of performance of 10 activities: (a) writing checks, paying bills, and keeping financial records; (b) assembling tax records and making out business and insurance papers; (c) shopping alone for clothes, household necessities, and groceries; (d) playing a game of skill such as bridge, other card game, or chess; (e) heating water for coffee or tea and turning off the stove; (f) preparing a balanced meal; (g) keeping track of current events; (h) paying attention to and understanding a TV program, book, or magazine; (i) remembering appointments, family occasions, and medications; and (j) travel out of the neighborhood. For each activity, four levels ranging from dependence to independence were presented. Two additional choices for activities not normally performed by the participant asked whether the participant was capable of performing the task now if necessary or would probably have difficulty. The four levels of function for each activity were assigned arbitrary weights as follows: dependent = 3, requires assistance = 2, has difficulty but does by self = 1, normal = 0. "Never did, but

could do now" was assigned a value of 0 and "never did and would have difficulty now," a value of 1. Total scores for the instrument were simple sums of scores for the individual activities.

FAQ data were not obtained for early participants. A similar, but less reliable instrument, was modified and expanded to the format described. Data are presented below for all 178 individuals for whom an alternate informant had completed the IADL or the FAQ. No conclusion differed when the analysis was repeated with the sample restricted to those on whom both IADL and FAQ scores were available.

RESULTS

Interrater reliability and validity of the Scale of Functional Capacity. — In their assessment of "residual mental function," the reviewing and examining neurologist showed remarkable agreement ($r = .97, p < .001$). The correlation between nurse's and examining neurologist's ordinal estimates of functional level was also high ($r_b = .802, p < .001$). These correlations may be an overestimate of agreement because the range of functional level among participants was restricted by recruitment of normal to mildly affected individuals.

A comparison of level assigned on the Scale of Functional Capacity by the examining neurologist with score on the IADL measure of social (in)dependence indicates a consistent trend toward greater dysfunction at higher ordinal levels (Table 2). Because the opinion of the alternate informant, but not the FAQ or IADL score, was available to the neurologist in an estimated 25% of interviews, these judgments

Table 2. Measures of Social Function and Cognitive Status by Categories of Functional Capacity^a

Global estimate of examining neurologist	Social function scales		Cognitive measures		
	FAQ ^b (n = 158)	IADL ^b (n = 178)	MMST (n = 195)	SDMT (n = 195)	Raven (n = 195)
Normal	3.42 ± 4.20 (104)	9.14 ± 2.37 (112)	29.01 ± 1.36 (124)	38.05 ± 8.30 (124)	7.90 ± 2.40 (124)
Questionably affected	12.00 ± 9.13 (15)	12.00 ± 4.64 (19)	27.95 ± 1.88 (21)	26.52 ± 9.67 (21)	4.71 ± 2.59 (21)
Mildly affected	14.63 ± 7.15 (16)	12.38 ± 4.47 (21)	24.13 ± 4.67 (23)	20.52 ± 8.32 (23)	5.30 ± 1.96 (23)
Moderately affected	22.90 ± 5.64 (20)	16.50 ± 3.41 (22)	19.78 ± 5.53 (23)	17.74 ± 12.52 (23)	4.39 ± 2.23 (23)
Moderately severely affected	29.00 ± 1.00 (3)	19.00 ± 4.58 (3)	19.00 ± 10.58 (3)	10.00 ± 9.54 (3)	5.67 ± 5.51 (3)
Severely affected	— ^c	23.00 — (1)	5.00 ± — (1)	7.00 ± — (1)	5.00 ± — (1)
Severely incapacitated	—	—	—	—	—

^a Results are expressed as means ± standard deviations. Numbers in parentheses denote number of participants for whom completed assessments were available. MMST, Mini-Mental State test; SDMT, Symbol Digit Modalities test.

^b Higher scores indicate greater disability.

^c Dashes indicate that no data are available or not applicable.

are not entirely independent. Scores on the brief cognitive function tests were, however, not available to the neurologist and indicate a similar trend.

Validity of the FAQ. — Alternate informant estimates of participant social function on the FAQ were highly correlated with their estimates on the IADL Scale ($r = .72$; 95% confidence interval estimate .64, .79). Using a multiple regression analysis to predict an individual's scores on various measures of cognitive and social function, alternate informant estimates on the FAQ and the IADL scales were entered as independent variables. Outcome variables so examined included scores on Mini-Mental State, Symbol Digit, Raven B, Mental Status Questionnaire, participant self-assessment of social function, global assessment of examining neurologist, consensus assessment of examining and reviewing neurologist, and the Mental Function Index derived from the Mini-Mental State, Symbol Digit, and Raven B combined. Scores on the FAQ always entered the regression equation first. Inclusion of IADL scores did not enhance the regression equation, as the increases in the multiple correlation coefficient (R) attributable to the IADL were close to zero. The largest single increase of .03 was obtained for prediction of participant self-assessment. By contrast, the FAQ explained an additional portion of the variance after prior entry of the IADL. Further contributions to R ranged from .04 to .17 with a median of .14 and a mean of .13. Coefficients of simple correlation between FAQ and various other measures of social and cognitive function were similar to or higher than the correlations between IADL score and these same measures (Table 3).

The neurologists' estimate of residual function

Table 3. Bivariate Correlations between Multiple Measures of Social and Cognitive Function^a

Measure	FAQ ($n = 158$)	IADL ($n = 178$)
Alternate informant IADL	.72	
Self-assessment social function	.79	.75
Global estimate both neurologists ^b	-.83	-.68
Mental function index	.75	.63
Mini-Mental State	-.71	-.55
Symbol Digit Modalities test	-.68	-.57
Raven, subtest B	-.41	-.42
Mental Status Questionnaire	-.76	-.62

^a Numbers in parentheses indicate number of participants.

^b Percentage of residual function.

Table 4. Correlation of Individual Functional Activities Questionnaire Items with Summary Estimates of Social Function

Item	FAQ total	IADL	Neurologist global estimate ^a
Writing checks, paying bills, balancing checkbook	.89	.60	-.72
Assembling tax records, business affairs, or papers	.83	.52	-.67
Shopping alone for clothes, household necessities, and groceries	.90	.71	-.73
Playing a game of skill, working on a hobby	.82	.63	-.65
Heat water, make a cup of coffee, turn off stove	.80	.68	-.65
Prepare a balanced meal	.86	.67	-.67
Keep track of current events	.81	.52	-.70
Pay attention to, understand, discuss TV, book, magazine	.86	.58	-.81
Remember appointments, family occasions, holidays, medications	.82	.56	-.70
Travel out of neighborhood, driving, arranging to take busses	.84	.64	-.68

^a Percentage of residual function.

correlated more highly with the FAQ ($r = -.83$) than the IADL ($r = -.68$) estimates by the alternate informant. As indicated above, statements of statistical significance can not be made with precision, because the neurologist interviewed some alternate informants, although most completed the questionnaire and returned it by mail. When the distribution of FAQ scores and IADL scores is reviewed by functional capacity level, there are greater differences among levels in the FAQ scores than the IADL scores and the degree of linear decline with level is greater for the FAQ (Table 2).

Comparison of scores on individual items with total scores on the IADL and FAQ scales and with the global estimate by the neurologist did not identify particularly poor items (Table 4). FAQ items displayed correlations of .80 to .90 with total FAQ scores. The range of item correlations with the total IADL score was .52 to .71, and with the neurologist assessment $-.65$ to $-.81$. Independence in shopping was the best single predictor, followed closely by writing checks and balancing a checkbook and by paying attention to TV or getting something out of a book or magazine. It is not surprising that the IADL did not correlate as highly with items mentioning business affairs or current events, as

Table 5. Estimates of Social Function among Older Adults by Diagnostic Category^a

Source and social function measure	Criterion diagnosis ^b				
	Normal	Depression only	Senile dementia	Senile dementia and depression	Other dementing illness
Self-assessment of activities	2.4 ± 9.5 (97)	9.0 ± 17.6 (28)	48.8 ± 36.3 (53)	46.4 ± 34.7 (10)	67.9 ± 29.0 (7)
Alternate informant, FAQ	2.8 ± 3.5 (80)	5.4 ± 5.5 (25)	18.5 ± 8.4 (39)	15.9 ± 7.0 (8)	18.7 ± 12.0 (6)
Alternate informant, IADL	8.9 ± 2.3 (88)	9.9 ± 2.6 (25)	14.3 ± 4.6 (48)	13.5 ± 5.4 (10)	14.7 ± 5.9 (7)

^a Results are means scores ± standard deviations; higher scores indicate more severe dysfunction. Numbers in parentheses refer to number of individuals for whom data were available.

^b Consensus diagnosis of examining and reviewing neurologist.

these activities are not sampled in the IADL scale.

FAQ, IADL, and self-assessment scores are indicated for the diagnostic categories identified by the consensus of examining and reviewing neurologist in Table 5. Mean scores of all three instruments differed significantly between demented individuals considered as a group (senile dementia, senile dementia and depression, other dementing illness) and persons not so affected (depression only or normal).

Classification of a person as dysfunctional when he or she was described on the FAQ by the alternate informant as dependent in more than two activities indicates the diagnostic utility of the instrument. With the stated criterion level, sensitivity was .85 and specificity .81. Similar classification utilizing IADL scores and the three-activity criterion resulted in a sensitivity of .57 and specificity of .92.

When used in combination with the mental function index, the FAQ score was helpful in distinguishing between normal and questionably demented older adults. Persons in the questionable range of mental function index values were classified as demented if described as (a) totally dependent on another to perform at least one activity and requiring advice or assistance with a second, or (b) requiring advice or assistance with three activities, or (c) some similar combination producing a disability score of 5 or more. This additional test reduced false-positive results from 25 to 14 and increased specificity of the mental function index from .80 to .89. This was achieved with an increase in the number of false-negative results from 5 to 6 with a decrease in sensitivity from .93 to .91.

DISCUSSION

The Functional Capacity Scale presented is sufficiently simple that it was readily adopted by another neurologist after joint discussion of three or four case abstracts and by two experi-

enced nurses after some two to three afternoons of utilizing the scale. Interrater reliability among neurologists was high ($r = .97$). Since the scale was designed to express severity of dementia, our efforts derive construct validity from the trend of scores on the individual cognitive function tests (Table 2). Because the scale levels are defined in terms of social function, still further assurance of construct validity derives from the observed progressive increase in IADL scores at each successive level (Table 2).

The present data confirm the concurrent validity of a new social function instrument, the FAQ, when compared with the IADL on a sample composed of clinically normal or mildly to moderately demented older adults. The FAQ also compares favorably with the IADL Scale in predicting an experienced neurologist's assessment, participants' own assessment of social function, and score on multiple brief tests of cognitive function.

The FAQ described in the present report is more sensitive than the established IADL Scale, as hoped from the selection of additional and more complex activities for query and the wider range of scale scores. Achievement of added sensitivity without substantive loss of specificity greatly enhances the value of this instrument for the target population: older adults in the community who are normal or only mildly dysfunctional. Because sample recruitment deliberately excluded most severely affected cases of senile dementia and persons suffering from other types of dementing illness, applicability of the FAQ to these groups will require further evaluation.

In its present state, the FAQ is of value as a screening tool for social dysfunction when used alone. It is of even more value in population studies to evaluate borderline cognitive function scores and reduce the number of false-positive results, so fewer individuals need be subjected to extended evaluation.

Additional work is in progress to improve the

format, clarify the phraseology, and measure the affective status and reliability of the alternate informant completing the FAQ. In a subsequent version, we have added probes to clarify and confirm the forced-choice segment of the instrument and to obtain further data on the onset and course of any dementing illness. Other areas for future research include broadening the scope of activities sampled and evaluation of the equivalence and/or predictive ability of individual activities.

Measurement of social function is critical in identifying those organic and affective states that are sufficiently serious to require therapeutic intervention. Prediction of outcome and serial evaluation of therapy are additional goals. Such measurements are essential to a further understanding of the relation between the cognitive functions measured by conventional neuropsychological tests and the everyday skills essential to daily living. The preliminary studies described suggest that these goals are attainable, although perhaps not yet achieved, by those willing to reopen this vital field of investigative endeavor.

REFERENCES

- Folstein, M. E., Folstein, S. W., & McHugh, P. R. 'Mini-Mental State': A practical method for grading the cognitive state of patients for the clinician. *Journal of Psychiatric Research*, 1975, 12, 189-198.
- Gallagher, D., Thompson, L. W., & Levy, S. M. Clinical psychological assessment of older adults. In L. W. Poon (Ed.), *Aging in the 1980s*. American Psychological Association, Washington, D.C., 1980.
- Gilleard, C. J., & Pattie, A. H. The Stockton Geriatric Rating Scale: A shortened version with British normative data. *British Journal of Psychiatry*, 1977, 131, 90-94.
- Kahn, R. L., Golfarb, A. I., Pollack, M., & Peck, A. Brief objective measures for the determination of mental status in the aged. *American Journal of Psychiatry*, 1960, 117, 326-328.
- Kleban, M., Lawton, M. P., Brody, E. M., & Moss, M. Behavioral observations of mentally impaired aged: Those who decline and those who do not. *Journal of Gerontology*, 1976, 31, 333-339.
- Kuriansky, J., & Gurland, B. The performance test of activities of daily living. *International Journal of Aging and Human Development*, 1976, 7, 343-352.
- Lawton M. P., & Brody, E. M. Assessment of older people: Self-maintaining and instrumental activities of daily living. *Gerontologist*, 1969, 9, 179-186.
- Pfeffer, R. I., Kurosaki, T. T., Harrah, C., Chance, J. M., Bates, D., Detels, R., Filos, S., & Butzke, C. A survey diagnostic tool for senile dementia. *American Journal of Epidemiology*, 1981, 114, 515-527.
- Plutchik, R., Conte, H., Lieberman, M., Bakur, M., Grossman, J., & Lehrman, N. Reliability and validity of a scale for assessing the functioning of geriatric patients. *Journal of the American Geriatrics Society*, 1970, 18, 491-500.
- Raven, J. C. Guide to using the coloured progressive matrices. H. K. Lewis & Company, Ltd., London, 1956.
- Remington, M., & Tyre, P. G. The social functioning schedule: A brief semistructured interview. *Social Psychiatry*, 1979, 14, 151-157.
- Smith, A. Symbol Digit Modalities Test. Western Psychological Services, Los Angeles, California, 1973.
- Spitzer, R. L., & Endicott, J. Medical and mental disorder: Proposed definition and criteria. In R. L. Spitzer & D. F. Klein (Eds.), *Critical issues in psychiatric diagnosis*. Raven Press, New York, 1978.
- Weissman, M. M., Sholomskas, D., Pottinger, M., Prusoff, B. A., & Locke, B. Z. Assessing depressive symptoms in five psychiatric populations: A validation study. *American Journal of Epidemiology*, 1977, 106, 203-214.